

1. What degree of parallelism allows for the optimal throughput (total number of calls in a period of time)? Why do you think the component shows this performance characteristic?
 - a. The larger the degree of parallelism is, the larger the number of total calls becomes, however the time also increases. That happens because the threads start to compete for the CPU resources. The most optimal degree of parallelism for the best throughput is the number of available CPU cores, in my case it is 8.
2. What sleep interval allows for the optimal throughput?
 - a. The best sleep interval for the optimal throughput is 0. Because there would be no sleep, the threads would be able to issue requests continuously, which would increase the throughput.
3. What would the numbers show if the initial load only loaded 500 items, where the first 500 add() calls in the second part of the test when the cache is not completely full, yet? Why would/does the benchmark behave this way? Hint: try loading only 500 items to see the actual results.
 - a. If the initial load only loads 500 items instead of the full cache size, the first 500 add() calls in the second part of the benchmark will just insert new items without removing anything. Because of that, the operations will be faster at the beginning since there is no eviction happening yet. Once the cache becomes full, the add() calls will start removing the oldest items, which makes them slower and changes the behavior of the benchmark.